Adaptive Dual-Arm Robotic Systems for Selective Electronics Disassembly

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1 Introduction

Electronics recycling currently employs a hybrid model. Manual dismantling is reserved for high-complexity devices like smartphones to isolate hazardous components (e.g., Li-ion batteries) and high-value PCBs . Bulk processing relies on semi-automated industrial shredding, magnetic separation, and near-infrared (NIR) sorting.

However, manual labor is slow and hazardous, while mechanical shredding results in "downcycling," pulverizing components and preventing the recovery of pure rare earth elements.

To advance the circular economy and meet WEEE Directive mandates, high-precision automation is essential. It mitigates the risks of hazardous substance release (e.g., lead, mercury) and enables the standardized recovery of critical materials lost during shredding, offering a scalable solution to the exponential growth of global e-waste.

2 The Collaborative Robotic Ecosystem

The system employs two 6-DOF collaborative robots (e.g., UR5e) integrated with ROS2 for synchronized dual-arm operations.

- **Workspace**: A fixed coordinate system maps the workbench, racks, and bins with sub-millimeter precision.
- **Safety**: Fenceless operation is achieved via internal force-limiters and external laser scanners that pause motion during human interaction.

3 Sensor Stack Integration

The system utilizes a diverse array of sensors listed in table 1 to navigate variable e-waste layouts.

Table 1: Sensor Configuration and Functions

Sensor Type	Primary Function
Global Camera	ID, pose estimation, & path planning.
RGB-D Cameras	3D mapping of micro-features.
Thermal Imaging	Detects battery thermal risks.
LiDAR/Proximity	Collision prevention (Human/Robot).
Piezoelectric	Tactile sensitivity for fragile parts.

4 Multi-Functional Pluggable Gripper

The primary end-effector for both robotic arms is a hybrid pluggable gripper. This design features a five-finger configuration that serves as a permanent grasping tool for component handling. However, the system is engineered with an internal mechanical interface: when the fingers open to their maximum aperture, a central coupling mechanism is exposed, allowing for the rapid attachment of auxiliary tools. This grippable tool architecture ensures that both arms can switch between handling parts and executing technical tasks without requiring a full end-effector swap.

4.1 Tool Categorization

- **Disassembly**: Includes torque-controlled electric screwdrivers, micro-spindles for milling stripped screws, and vacuum tips for screen/backplate removal.
- **Triage & Safety**: Features conductive probe grippers for PCB testing, diagnostic equipment handlers, and insulated tips for safe battery extraction.

5 System Intelligence and Compute Architecture

Processing variable e-waste, where every device presents a unique internal state, demands a high-performance compute architecture. A high-performance edge computing node serves as the system's brain, orchestrating three critical data streams: **Real-time AI vision** for defect detection, **Sensor Fusion** (haptics & thermal) for safety overrides, and **ROS2-based collaborative control** for dual-arm synchronization.

5.1 Schematic Loading and Perception

Device schematics are retrieved from a centralized database to serve as digital twins. This data enables precise component localization (e.g., screws, batteries) and generates collision-free path planning for the dual-arm system.

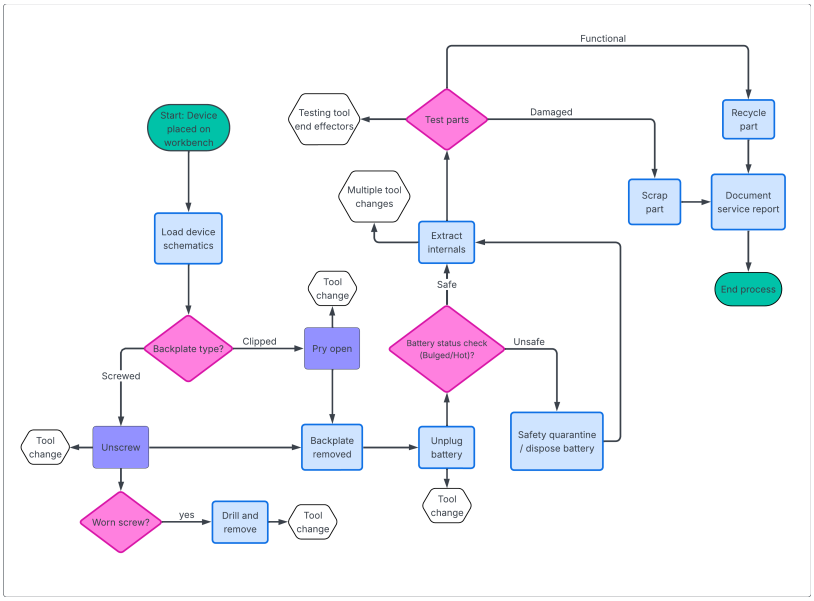

Figure 1: Robotic Electronics Disassembly Flowchart: Integrating Schematic Loading and Automated Triage.

Table 2: Comparative Performance Analysis

Metric	Manual Labor	Rigid Automation (Daisy)	Adaptive Dual-Arm (Target)
Throughput	≈10–20 phones/hr	≈200 iPhones/hr	≈120 phones/hr
Cycle Time	180–600s	18s	30s
Variety	High (Any form factor)	Low–Med (iPhones only)	High (Multi-brand via AI)
Purity	Variable	High	High (Sensor-verified)
Cost Efficiency	Low/Negative profit	High profit (Volume)	High profit (Flexibility)

6 Results and Comparison

The adaptive dual-arm system aims to bridge the gap between low-throughput manual labor and high-volume rigid automation. Table 2 highlights the key performance metrics. The system targets a throughput of approximately 120 phones per hour, significantly faster than manual labor while maintaining the flexibility to handle multiple brands, a key advantage over rigid systems like Apple's Daisy.

References

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